Software Development Models

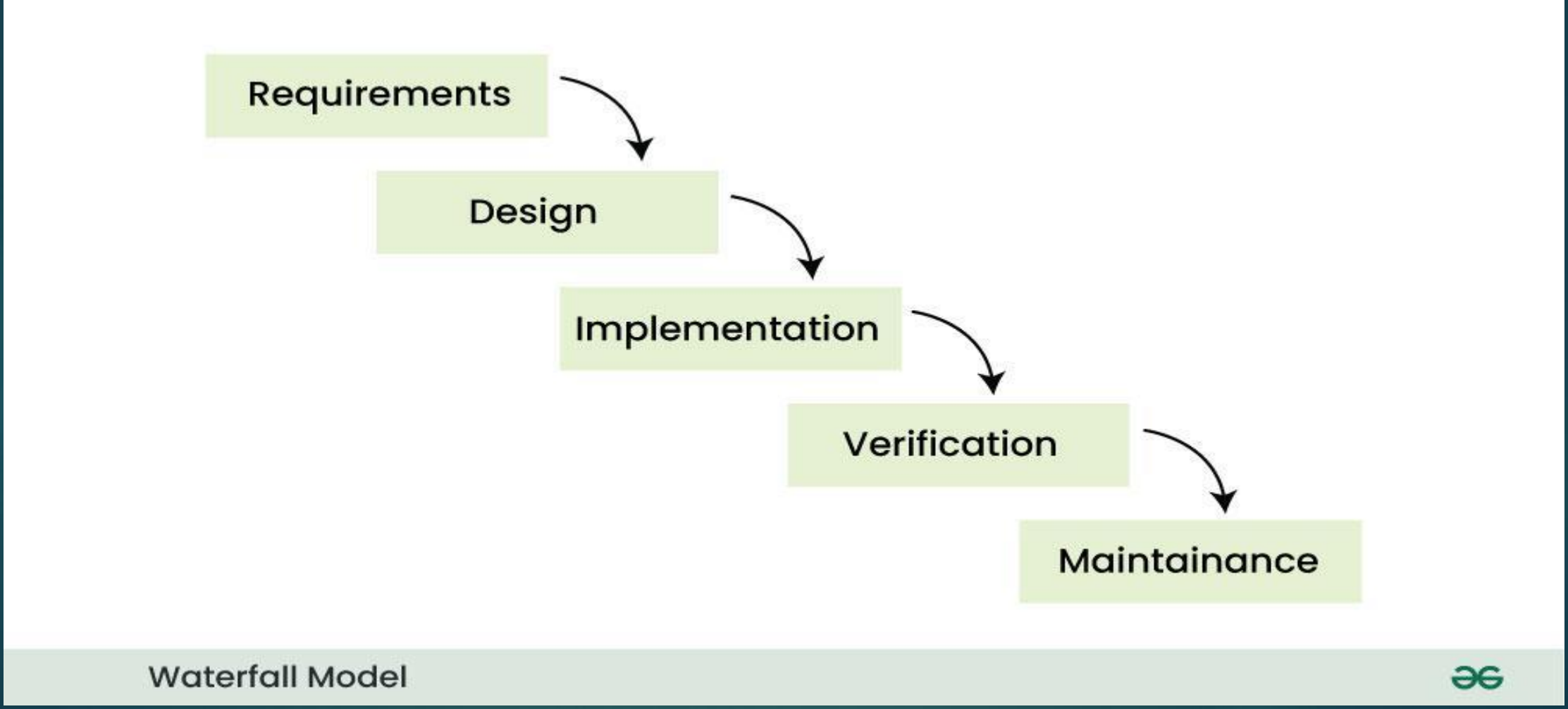
1. Waterfall Model
2. V-Model
3. Incremental Model
4. RAD Model
5. Iterative Model
6. Spiral Model
7. Prototype model
8. Agile Model

Why are companies shifting toward agile Software Development models?

Waterfall Model

Waterfall model is a famous and good version of SDLC(System Development Life Cycle) for software engineering. The waterfall model is a linear and sequential model, which means that a development phase cannot begin until the previous phase is completed. We cannot overlap phases in the waterfall model.

Waterfall Model



1. **Requirement phase:-** Requirement phase is the first phase of the waterfall model. In this phase the requirements of the system are collected and documented. This phase is very crucial because the next phases are based on this phase.
2. **Design phase:-** Design phase is based on the fact how the software will be built. The main objective of the design phase is to prepare the blueprint of the software system so that no problems are faced in the coming phases and solutions to all the requirements in the requirement phase are found.
3. **Implementation phase:-** In this phase, hardware, software and application programs are installed and the database design is implemented. Before the database design can be implemented, the software has to go through a testing, coding, and debugging process. This is the longest lasting phase in waterfall.

4. **Verification phase:-** In this phase the software is verified and it is evaluated that we have created the right product. In this phase, various types of testing are done and every area of the software is checked. It is believed that if we do not verify the software properly and there is any defect in it then no one will use it, hence verification is very important. One advantage of verification is that it reduces the risk of software failure.
5. **Maintenance phase:-** This is the last phase of waterfall. When the system is ready and users start using it, then the problems that arise have to be solved time-to-time. Taking care of the finished software and maintaining it as per time is called maintenance.

Advantages of the Waterfall Model

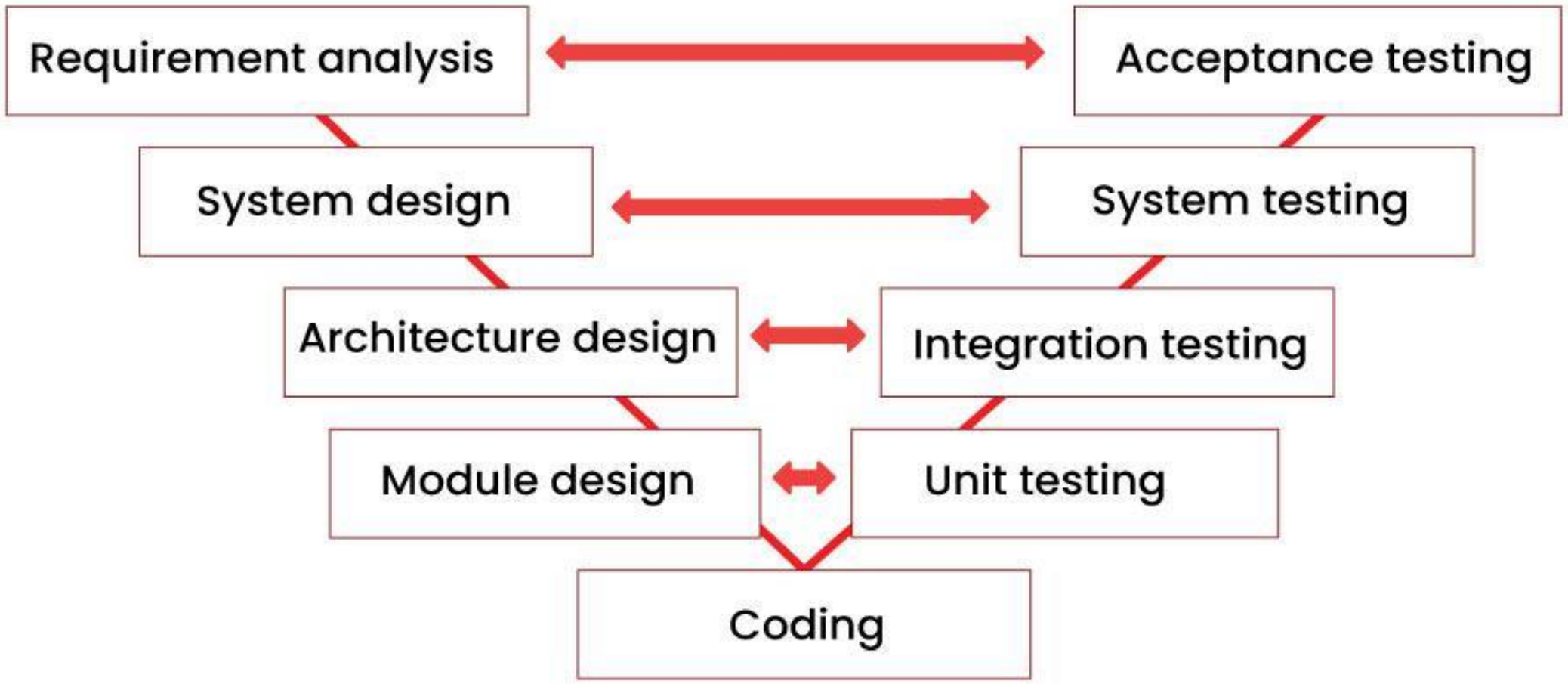
- ✓This model is simple and easy to understand.
- ✓This is very useful for small projects.
- ✓This model is easy to manage.
- ✓The end goal is determined early.
- ✓Each phase of this model is well explained.
- ✓It provides a structured way to do things.

Disadvantages of Waterfall Model

- ✓ In this model, complete and accurate requirements are expected at the beginning of the development process.
- ✓ Working software is not available for very long during the development life cycle.
- ✓ We cannot go back to the previous phase due to which it is very difficult to change the requirements.
- ✓ Risk is not assessed in this, hence there is high risk and uncertainty in this model.
- ✓ In this the testing period comes very late.
- ✓ Due to its sequential nature this model is not realistic in today's world.
- ✓ This is not a good model for large and complex projects.

2. V-Model

- V-Model is an SDLC model, it is also called Verification and Validation Model. V-Model is widely used in the software development process, and it is considered a disciplined model. In V-Model, the execution of each process is sequential, that is, the new phase starts only after the previous phase ends.
- It is based on the association of testing phase with each development phase that is in V-Model with each development phase, its testing phase is also associated in a V-shape in other words both software development and testing activities take place at the same time.


Requirement analysis
Acceptance testing
System design
System testing
Architecture design
Integration testing
Module design
Unit testing
Coding

V-Model

1. **Requirements analysis:-** This is the first phase of the development cycle, in which the requirements of the product are analyzed according to the customer's needs. In this phase, product related requirements are thoroughly collected from the customer. This is a very important phase because this phase determines the coming phases. In this phase, acceptance tests are designed for later use.
2. **System design:-** When we have the requirements of the product, after that we prepare a complete design of the system. In this phase, a complete description of the hardware and all the technical components required to develop the product .
3. **Architectural design:-** In this phase architectural specifications are designed. It contains the specification of how the software will link internally and externally with all the components. Therefore this phase is also called high level design (HLD).

Validation Phases of V-Model

1. **Unit testing:-** In the unit testing phase, the unit tests created during the module design phase are executed. Unit testing is code level testing, it only verifies the technical design. Therefore it is not able to test all the defects.
2. **Integration testing:-** In integration testing, the integration tests created in the architectural design phase are executed. Integration testing ensures that all modules are working well together.
3. **System testing:**– In system testing, the system tests created in the system design phase are executed. System tests check the complete functionality of the system. In this, more attention is given to performance testing and regression testing.
4. **Acceptance testing:-** In acceptance testing, the acceptance tests created in the requirement analysis phase are executed. This testing ensures that the system is compatible with other systems. And in this, non-functional issues like:- load time, performance etc. are tested in the user environment.

Advantages of V-Model

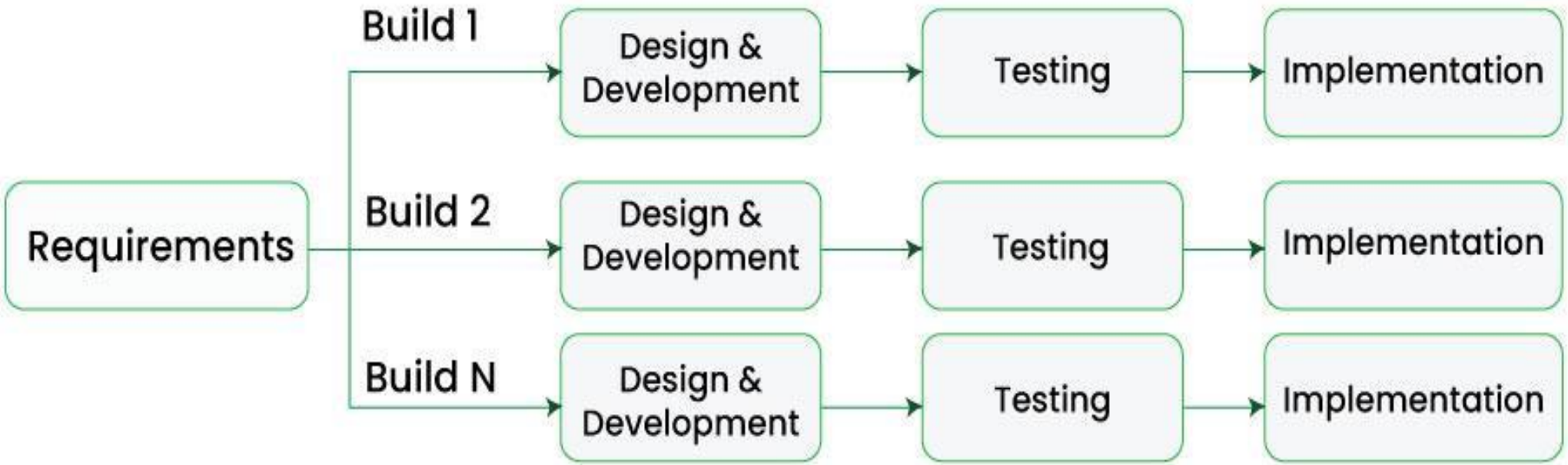
- ✓ This is a simple and easy to use model.
- ✓ Planning, testing and designing tests can be done even before coding.
- ✓ This is a very disciplined model, in which phase-by-phase development and testing goes on.
- ✓ Defects are detected in the initial stage itself.
- ✓ Small and medium-scale developments can be easily completed using it.
- ✓ Easy-to-use

Disadvantages of V-Model

- ✓ This model is not suitable for any complex projects.
- ✓ There remains both high risk and uncertainty.
- ✓ This is not a suitable model for an ongoing project.
- ✓ This model is not at all suitable for a project which is unclear and in which there are changes in the requirement.

3. Incremental Model

- In Incremental Model, the software development process is divided into several increments and the same phases are followed in each increment. In simple language, under this model, a complex project is developed in many modules or builds.
- For example, we collect the customer's requirements, now instead of making the entire software at once, we first take some requirements and based on them create a module or function of the software and deliver it to the customer. Then we take some more requirements and based on them add another module to that software.
- Similarly, modules are added to the software in each increment until the complete system is created. However, the requirements for making a complex project in multiple iterations/parts should be clear.



Incremental Model

Following are the different phases of Incremental Model:-

1. **Communication**: In the first phase, we talk face to face with the customer and collect his mandatory requirements. Like what functionalities does the customer want in his software, etc.
2. **Planning**: In this phase the requirements are divided into multiple modules and planning is done on their basis.
3. **Modeling**: In this phase the design of each module is prepared. After the design is ready, we take a particular module among many modules and save it in DDS (Design Document Specification). Diagrams like ERDs and DFDs are included in this document.

4. Construction: Here we start construction based on the design of that particular module. That is, the design of the module is implemented in coding. Once the code is written, it is tested.

5. Deployment: After the testing of the code is completed, if the module is working properly then it is given to the customer for use. After this, the next module is developed through the same phases and is combined with the previous module. This makes new functionality available to the customer. This will continue until complete modules are developed.

Advantages of Incremental Model

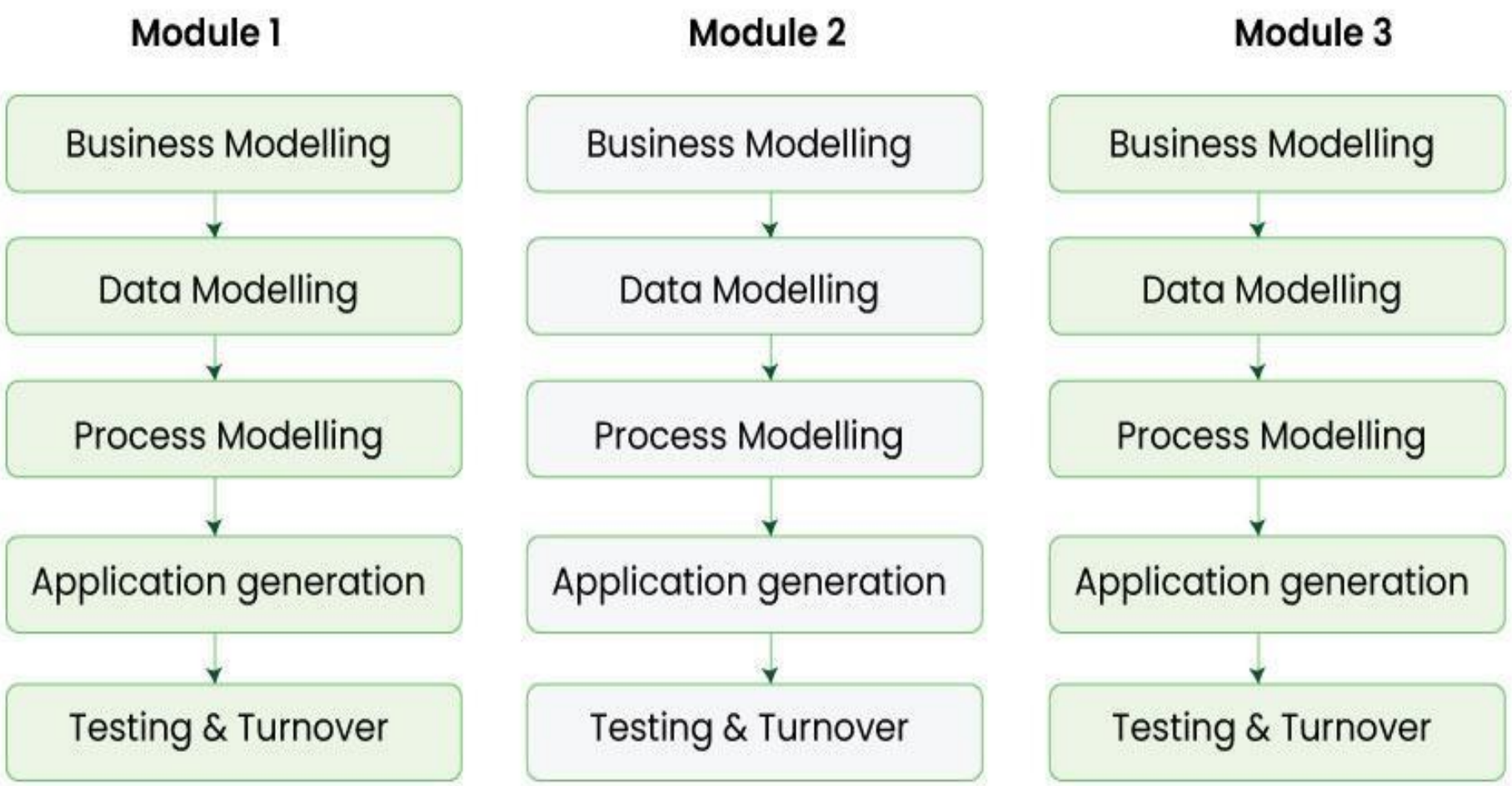
- ✓ Important modules/functions are developed first and then the rest are added in chunks.
- ✓ Working software is prepared quickly and early during the software development life cycle (SDLC).
- ✓ This model is flexible and less expensive to change requirements and scope.
- ✓ The customer can respond to each module and provide feedback if any changes are needed.
- ✓ Project progress can be measured.
- ✓ It is easier to test and debug during a short iteration.
- ✓ Errors are easy to identify.

Disadvantages of the Incremental Model

- ✓ Management is a continuous activity that must be handled.
- ✓ Before the project can be dismantled and built incrementally,
- ✓ The complete requirements of the software should be clear.
- ✓ This requires good planning and designing.
- ✓ The total cost of this model is higher.

4. RAD Model

- RAD model stands for rapid application development model. The methodology of RAD model is similar to that of incremental or waterfall model. It is used for small projects.
- If the project is large then it is divided into many small projects and these small projects are planned one by one and completed. In this way, by completing small projects, the large project gets ready quickly.
- In RAD model, the project is completed within the given time and all the requirements are collected before starting the project. It is very fast and there are very less errors in it.


Module 1
Business Modelling
Data Modelling
Process Modelling
Application generation
Testing & Turnover
Module 2
Business Modelling
Data Modelling
Process Modelling
Application generation
Testing & Turnover
Module 3
Business Modelling
Data Modelling
Process Modelling
Application generation
Testing & Turnover

RAD Model

The following phases are followed in RAD model:-

1. **Business modeling:** In this phase, the business model is designed on the basis of whatever functions the business has. If we speak in a little technical language, then we design the business model for the product on the basis of flow of information and distribution of information between different business channels. Here information flow means what type of information drives the business, where the information comes from and where it goes, who generates it, etc. This means that a complete business analysis is done in this phase.
2. **Data modeling:** Using the business model we had prepared, the data objects required for the business are defined.
3. **Process modeling:** The data objects that we defined in the data modeling phase are converted to establish the business information flow. It is necessary to achieve specific business objectives.

4. **Application generation:** In this phase we start building the software based on the output of the above three phases. For this we take the help of automation tools. However, in this phase we do not develop the actual software but make a working prototype.
5. **Testing and turnover:** Whatever prototype we have prepared or whatever components and interfaces we have, they are tested in this phase. Since prototypes are tested separately during each iteration, the overall testing time in rapid application development is reduced.

Advantage of RAD Model:-

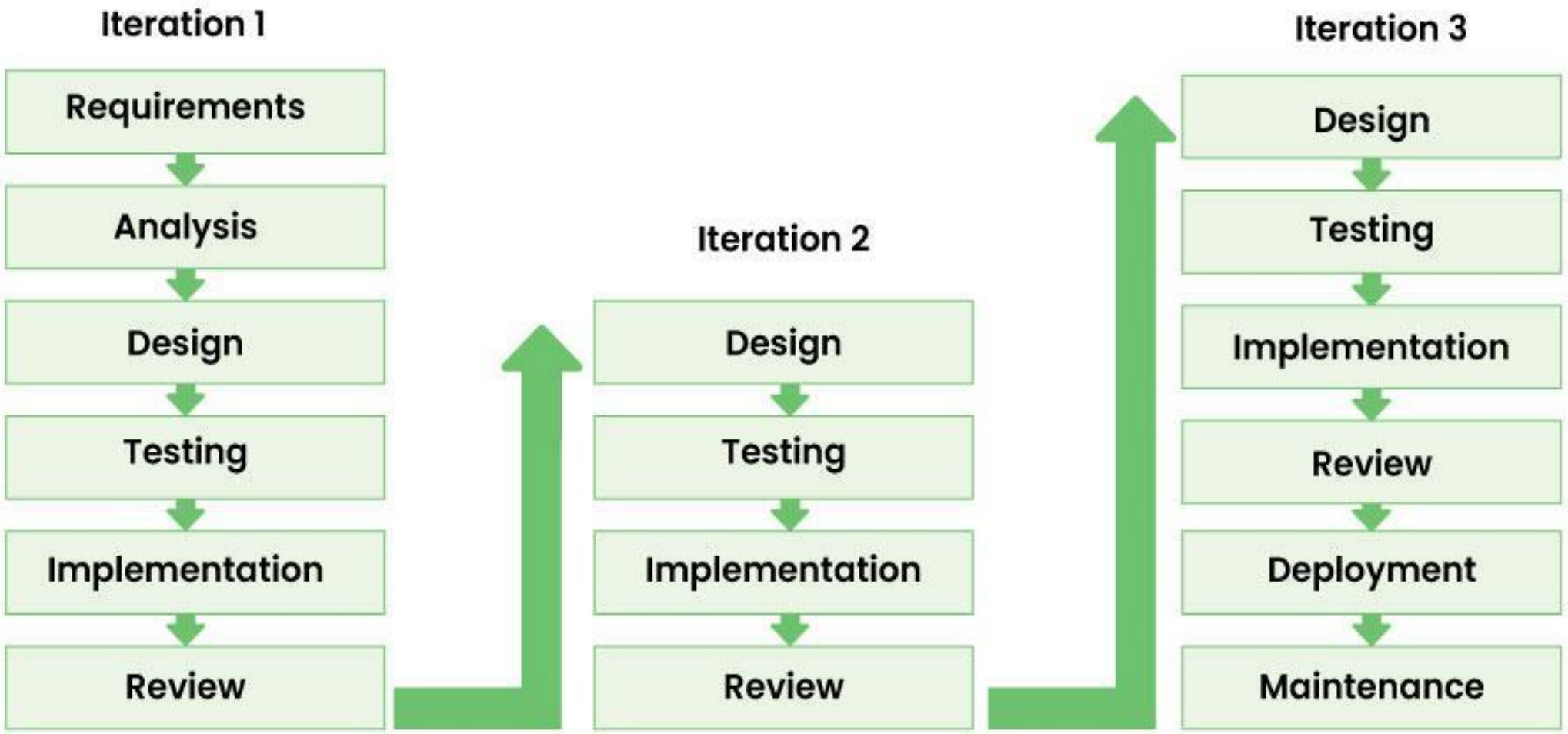
- ✓ It reduces the time taken in development.
- ✓ In this the components are reused.
- ✓ It is flexible and it is easy to make any changes in it.
- ✓ It is easy to transfer like scripts because high level abstraction and intermediate codes are used in it.
- ✓ There are very few defects in it because it is a prototype by nature.
- ✓ In this, productivity can be increased in less time with less people.
- ✓ It is cost effective.
- ✓ It is suitable for small projects.

Disadvantages of RAD Model:-

- ✓ In this we need highly skilled developers and designers.
- ✓ It is very difficult to manage.
- ✓ It is not suitable for project that are complex and takes long time.
- ✓ In this, feedback from the client is required for the development of each phase.
- ✓ Automated code generation is very expensive.
- ✓ This model is suitable only for component based and scalable systems.

5. Iterative Model

- In Iterative model we start developing the software with some requirements and when it is developed, it is reviewed. If there are requirements for changes in it, then we develop a new version of the software based on those requirements. This process repeats itself many times until we get our final product.
- So, in Iterative model a software is developed by following several iterations. Iteration means that we are repeating the development process again and again



Iterative model

The different phases of iterative model are as follows:-

1. **Requirement gathering & analysis:** In this phase, all the software requirements of the customer are collected and it is analyzed whether those requirements can be meet or not. Besides, it is also checked whether this project will not go beyond our budget.
2. **Design**: In this phase the design of software is prepared. For this, various diagrams like Data Flow diagram, class diagram, activity diagram, state transition diagram, etc. are used.
3. **Implementation**: Now the design of software is implemented in coding through various programming languages. We also call this coding phase.
4. **Testing**: After the coding of the software is done, it is now tested so that the bugs and errors present in it can be identified. To do this, various testing techniques like performance testing, security testing, requirement testing, stress testing, etc. are done.

5. **Deployment**: Finally the software is given to the customer. After this the customer starts using that software in his work environment.
6. **Review**: After the software is deployed in its work environment, it is reviewed. If any error/bug is found or any new requirements come in front of developer, then again these phases are repeated with new iteration and a new version is developed.
7. **Maintenance**: In this phase we look at customer feedback, solve problems, fix errors, update software, etc.

Advantage of Iterative model:-

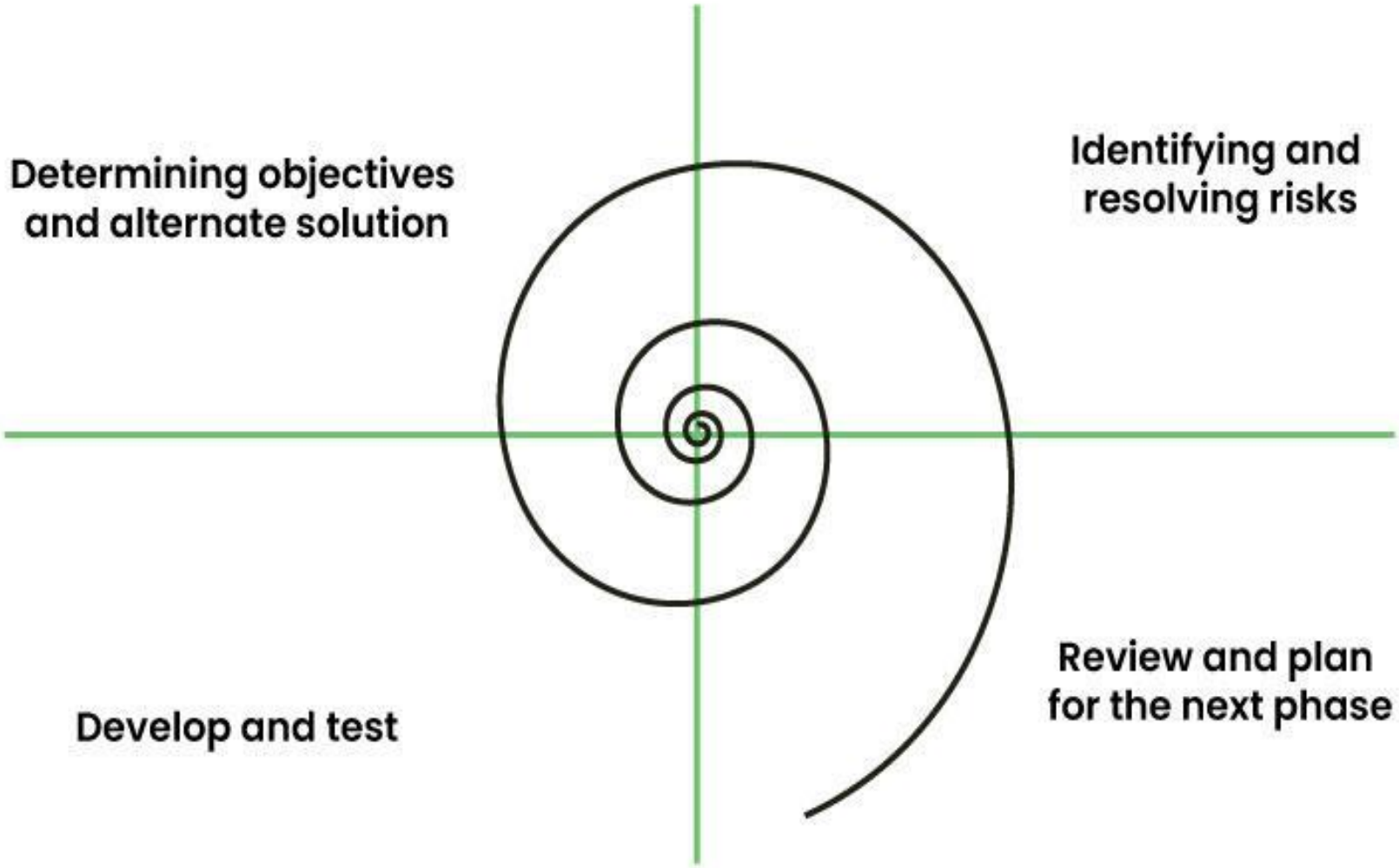
- ✓ In iterative models, bugs and errors can be identified quickly.
- ✓ Under this model, software is prepared quickly with some specifications.
- ✓ Testing and debugging the software becomes easier during each iteration.
- ✓ We get reliable feedback from users along with blueprints.
- ✓ This model is easily adaptable to constantly changing needs.
- ✓ During the software development process, additional time is devoted to development and limited time to documentation.
- ✓ Risks are identified and resolved during iteration.

Disadvantage of Iterative model:-

- ✓ Iterative model is not suitable for small projects.
- ✓ Since we have to repeat iterations many times in the software development process due to which we require more resources.
- ✓ Since the requirements are constantly changing, we have to make frequent changes in the software.
- ✓ Due to constantly changing requirements, the budget of the project also increases and it takes more time to complete it.
- ✓ In this model, it is complicated to control the entire process of software development.
- ✓ It is very difficult to tell by what date the complete software will be ready.

6. Spiral Model

- Spiral model is a software development process model. This model has characteristics of both iterative and waterfall models. This model is used in projects which are large and complex. This model was named spiral because if we look at its figure, it looks like a spiral, in which a long curved line starts from the center point and makes many loops around it. The number of loops in the spiral is not decided in advance but it depends on the size of the project and the changing requirements of the user. We also call each loop of the spiral a phase of the software development process.
- A software project goes through these loops again and again in iterations. After each iteration a more and more complete version of the software is developed. The most special thing about this model is that risks are identified in each phase and they are resolved through prototyping. This feature is also called Risk Handling.



Spiral model

Phases are as follows:-

1. **Determining objectives and alternate solutions:** In the first phase, whatever requirements the customer has related to the software are collected. On the basis of which objectives are identified and analyzed and various alternative solutions are proposed.
2. **Identifying and resolving risks:** In this phase, all the proposed solutions are assessed and the best solution is selected. Now that solution is analyzed and the risks related to it are identified. Now the identified risks are resolved through some best strategy.
3. **Develop and test:** Now the development of software is started. In this phase various features are implemented, that is, their coding is done. Then those features are verified through testing.
4. **Review and plan for the next phase:** In this phase the developed version of the software is given to the customer and he evaluates it. Gives his feedback and tells new requirements. Finally planning for the next phase (next spiral) is started.

Advantages of Spiral Model:

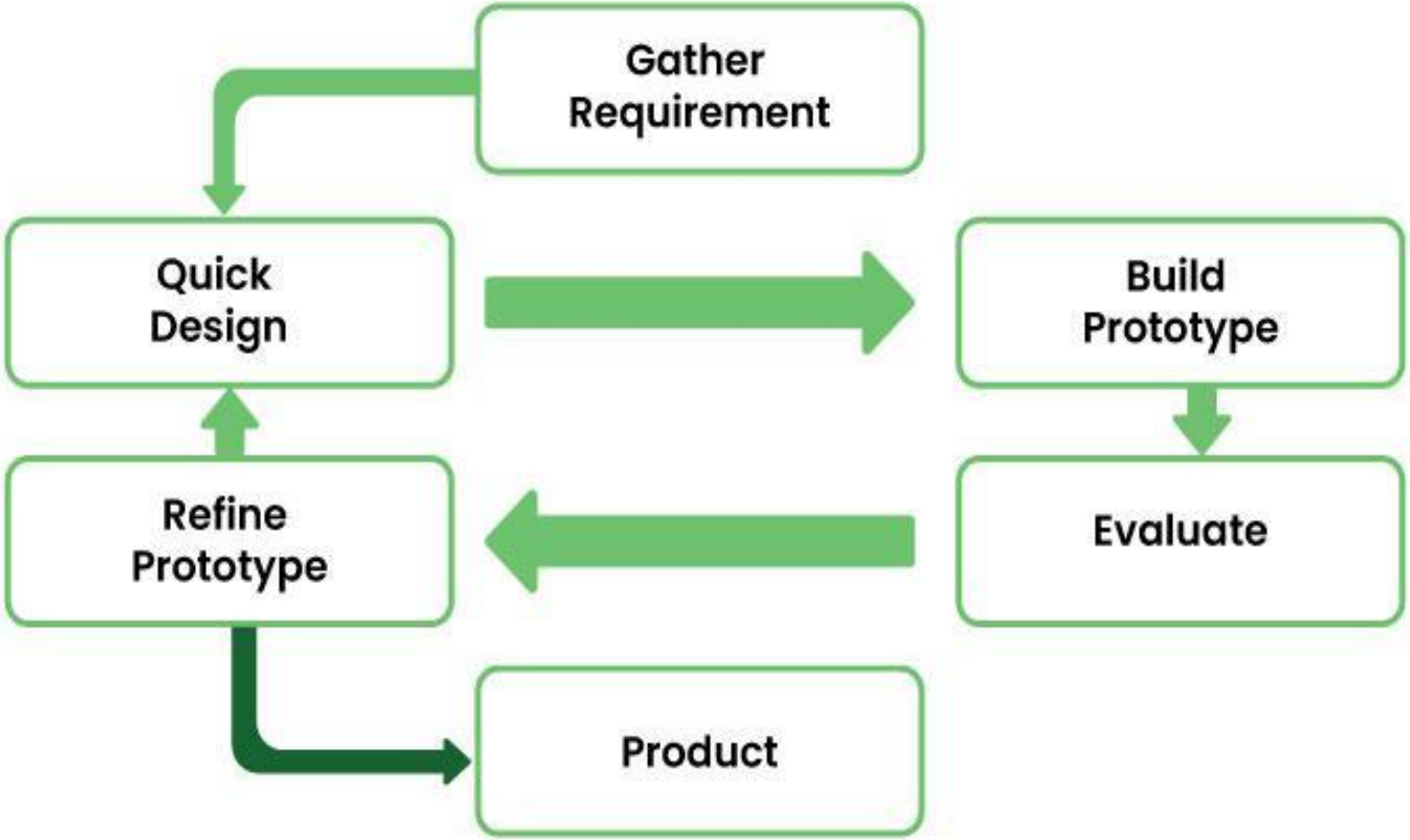
- ✓ If we have to add additional functionality or make any changes to the software, then through this model we can do so in the later stages also.
- ✓ Spiral model is suitable for large and complex projects.
- ✓ It is easy to estimate how much the project will cost.
- ✓ Risk analysis is done in each phase of this model.
- ✓ The customer can see the look of his software only in the early stages of the development process.
- ✓ Since continuous feedback is taken from the customer during the development process, the chances of customer satisfaction increases.

Disadvantage of Spiral Model:-

- ✓ This is the most complex model of SDLC, due to which it is quite difficult to manage.
- ✓ This model is not suitable for small projects.
- ✓ The cost of this model is quite high.
- ✓ It requires more documentation than other models.
- ✓ Experienced experts are required to evaluate and review the project from time to time.
- ✓ Using this model, the success of the project depends greatly on the risk analysis phase.

7. Prototype model

- Prototype model is an activity in which prototypes of software applications are created. First a prototype is created and then the final product is manufactured based on that prototype.
- ✓ The prototype model was developed to overcome the shortcomings of the waterfall model.
- ✓ This model is created when we do not know the requirements well.
- ✓ The specialty of this model is that this model can be used with other models as well as alone.



Prototype model

The prototype model has the following phases:-

1. **Requirement gathering:** The first step of prototype model is to collect the requirements, although the customer does not know much about the requirements but the major requirements are defined in detail.
2. **Build the initial prototype:** In this phase the initial prototype is built. In this some basic requirements are displayed and user interface is made available.
3. **Review the prototype:** When the construction of the prototype is completed, it is presented to the end users or customer and feedback is taken from them about this prototype. This feedback is used to further improve the system and possible changes are made to the prototype.
4. **Revise and improve the prototype:** When feedback is taken from end users and customers, the prototype is improved on the basis of feedback. If the customer is not satisfied with the prototype, a new prototype is created and this process continues until the customer gets the prototype as per his desire.

Advantages of Prototype model :-

- ✓ Prototype Model is suggested to create applications whose prototype is very easy and which always includes human machine interaction within it.
- ✓ When we know only the general objective of creating software, but we do not know anything in detail about input, processing and output. Then in such a situation we make it a Prototype Model.
- ✓ When a software developer is not very sure about the capability of an algorithm or its adaptability to an operating system, then in this situation, using a prototype model can be a better option.

Disadvantages of Prototype model :-

- When the first version of the prototype model is ready, the customer himself often wants small fixes and changes in it rather than rebuilding the system. Whereas if the system is redesigned then more quality will be maintained in it.
- Many compromises can be seen in the first version of the Prototype Model.
- Sometimes a software developer may make compromises in his implementation, just to get the prototype model up and running quickly, and after some time he may become comfortable with making such compromises and may forget that it is completely inappropriate to do so.

8. Agile Model

- Agile model is a combination of iterative and incremental models, that is, it is made up of iterative and incremental models.
- In Agile model, focus is given to process adaptability and customer satisfaction.
- The agile model was created mainly to make changes in the middle of software development so that the software project can be completed quickly.

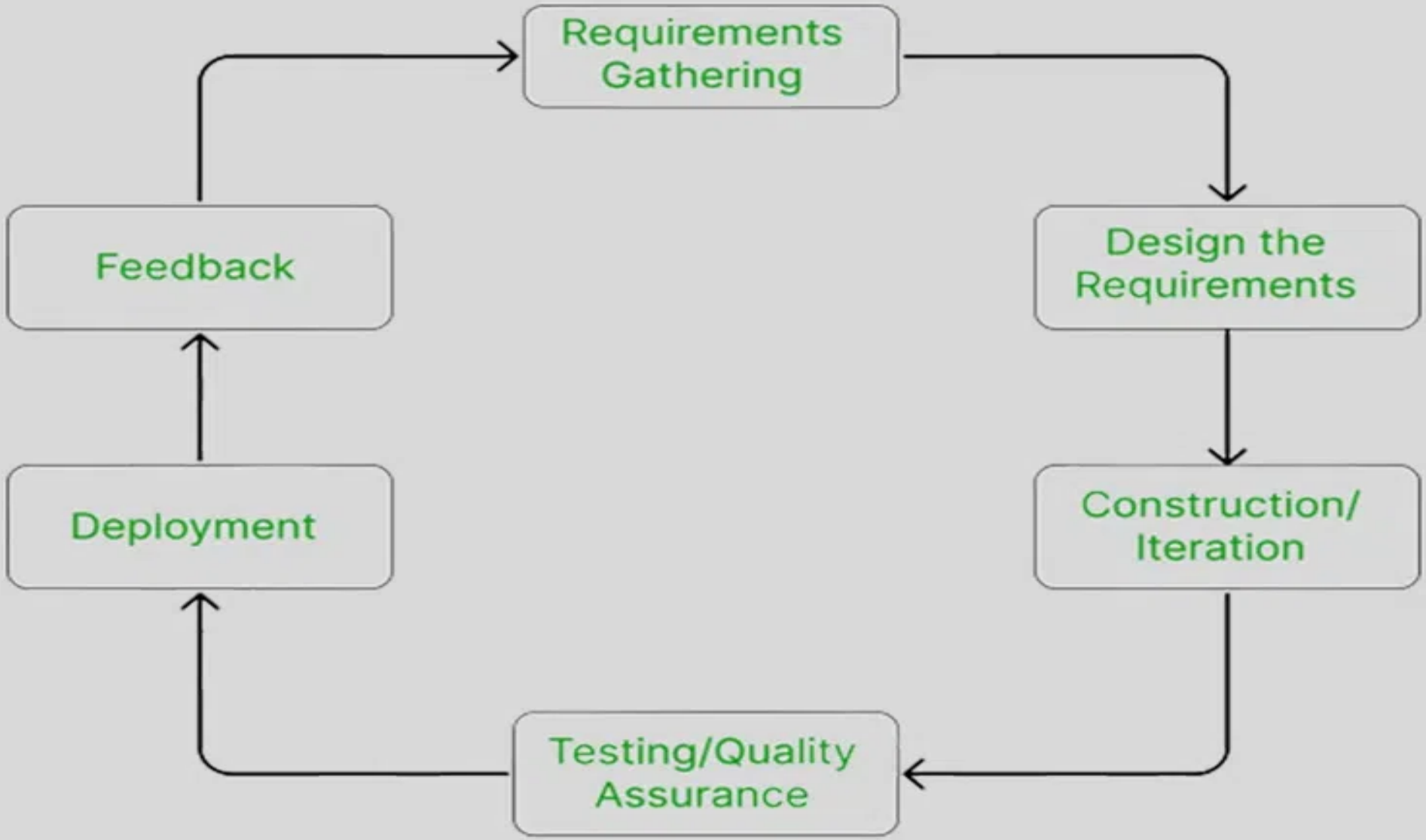


Fig:- Steps in Agile SDLC Model

Principles of Agile model:-

- There is a customer representative in the development team to maintain contact with the customer during software development and to understand the requirement. When an iteration is completed, stakeholders and customer representatives review it and re-evaluate the requirements.
- Demo of working software is given to understand the customer's requirements. That is, it does not depend only on documentation.
- Incremental versions of the software have to be delivered to the customer representative after a few weeks.

- In this model it is advised that the size of the development team should be small (5 to 9 people) so that the team members can communicate face to face.
- Agile model focuses on the fact that whenever any changes have to be made in the software, it should be completed quickly.
- In agile development, two programmers work together. One programmer does the coding and the other reviews that code. Both the programmers keep changing their tasks, that is, sometimes one does coding and sometimes someone reviews.

1. Requirement Gathering:- In this step, the development team must gather the requirements, by interaction with the customer. development team should plan the time and effort needed to build the project. Based on this information you can evaluate technical and economical feasibility.

2. Design the Requirements:- In this step, the development team will use user-flow-diagram or high-level UML diagrams to show the working of the new features and show how they will apply to the existing software. Wireframing and designing user interfaces are done in this phase.

3. Construction / Iteration:- In this step, development team members start working on their project, which aims to deploy a working product.

4. Testing / Quality Assurance:- Testing involves Unit Testing, Integration Testing, and System Testing. A brief introduction of these three tests is as follows:

5. Unit Testing:- Unit testing is the process of checking small pieces of code to ensure that the individual parts of a program work properly on their own. Unit testing is used to test individual blocks (units) of code.

- **Integration Testing:-** Integration testing is used to identify and resolve any issues that may arise when different units of the software are combined.
- **System Testing:-** Goal is to ensure that the software meets the requirements of the users and that it works correctly in all possible scenarios.

5. Deployment:- In this step, the development team will deploy the working project to end users.

6. Feedback:- This is the last step of the **Agile Model.** In this, the team receives feedback about the product and works on correcting bugs based on feedback provided by the customer.

Agile has the following models:-

1. Scrum
2. Kanban
3. Lean Software Development
4. SAFe (Scaled Agile Framework)
5. Extreme Programming (XP)

1. Scrum

Scrum is the type of Agile Framework, widely known for its incremental approach. It involves sprint cycles, daily stand-up meetings, and a focus, on product backlogs.

Features of Scrum in software development:

- Scrum is light-weighted framework
- Scrum emphasizes self-organization
- Scrum is simple to understand
- Scrum framework helps the team to work together

2. Kanban

Kanban focuses on visualizing the workflow, limiting work in progress, and optimizing work flow. It’s a framework for both technical and non-technical teams.

Features of Kanban in Software Development

- Kanban visualizes the workflow, It is crucial to visualize and make visible the workflow in order to understand how work is done.
- Kanban encourages small incremental, continuous, and evolutionary changes.
- Kanban sets limits on the number of tasks that can be worked on at the same time, promoting a steady controlled workflow.
- Kanban support early feedback from clients and the pull system are important

3. Lean Software Development

Lean software development is an agile framework that aim to eliminate waste improve efficiency and maximize the value delivery. They can be applied not to software development but across various industries.

Features of LSD in Software Development

- LSD removes the unnecessary process stages when designing software.
- LSD simplifies the development process and saves time by removing unnecessary stages.
- LSD reduces waste by minimizing unnecessary features.
- LSD enhance innovation through experimentation and creativity.

4. SAFe (Scaled Agile Framework)

SAFe is specifically designed for the use of organizations. It provides a framework for scaling practices to an enterprise level while ensuring both alignment and collaboration across teams.

Features of SAFe in Software Development

- SAFe promotes alignment between teams and individuals at all levels of the organization.
- SAFe accelerates product development and delivery.
- SAFe enhanced product quality
- SAFe enhance a culture of openness and visibility within organizations.

5. Extreme Programming (XP)

Extreme Programming is an agile framework that focuses on practices and it is particularly suitable for software development projects. It encourages integration, test-driven development and close collaboration, with customers

Features of Extreme Programming in Software Development

- The XP model is very useful in small projects.
- The XP model is well-suited for web development projects.
- The XP model is useful for collaborative projects
- The XP model can be used in projects that have a tight deadline.

Characteristics of the Agile Process

- ✓ Agile processes must be adaptable to technical and environmental changes. That means if any technological changes occur, then the agile process must accommodate them.
- ✓ The development of agile processes must be incremental. That means, in each development, the increment should contain some functionality that can be tested and verified by the customer.
- ✓ The customer feedback must be used to create the next increment of the process.
- ✓ The software increment must be delivered in a short span of time.
- ✓ It must be iterative so that each increment can be evaluated regularly.

When To Use the Agile Model?

- When frequent modifications need to be made, this method is implemented.
- When a highly qualified and experienced team is available.
- When a customer is ready to have a meeting with the team all the time.
- when the project needs to be delivered quickly.
- Projects with few regulatory requirements or not certain requirements.
- projects utilizing a less-than-strict current methodology
- Those undertakings where the product proprietor is easily reachable
- Flexible project schedules and budgets.

Advantages of Agile Model:-

- ✓ In this, two programmers work together due to which the code is error free and there are very few mistakes in it.
- ✓ In this the software project is completed in a very short time.
- ✓ In this the customer representative has an idea of each iteration so that he can easily change the requirement.
- ✓ This is a very realistic approach to software development.
- ✓ In this, focus is given on teamwork.
- ✓ There are very few rules in this and documentation is also negligible.
- ✓ There is no need for planning in this.
- ✓ It can be managed easily.
- ✓ It provides flexibility to developers.

Disadvantages of Agile Model:-

- It cannot handle complex dependencies.
- Due to lack of formal documentation in this, there is confusion in development.
- It mostly depends on the customer representative, if the customer representative gives any wrong information then the software can become wrong.
- Only experienced programmers can take any decision in this. New programmers cannot take any decision.
- In the beginning of software development, it is not known how much effort and time will be required to create the software.

Why companies are shifting toward agile Software Development models?

- In earlier times, iterative waterfall model was used to create software. But in today's time developers have to face many problems. The biggest problem is that in the middle of software development, the customer asks to make changes in the software. It takes a lot of time and money to make these changes. The agile model was created mainly to make changes in the middle of software development so that the software project can be completed quickly. Agile focuses on customer-centric approach that constantly take feedback from customer and make changes in the middle of software developer according to their needs and expectation. Agile Methodologies like scrum adopt iterative and incremental approach this accelerate the time-to-market for new features and product.